

Project 2: The Finite-Volume Method

AE 623, Computational Fluid Dynamics II, Winter 2026

Due: February 20, 11:59pm, electronically via Canvas

1 Problem Description

1.1 Introduction

In this project you will write first and second-order finite-volume solvers to perform simulations of inviscid, compressible flow through the turbine blade passage that you meshed in the last project. In this description, the “coarse” mesh refers to your locally-refined mesh of ~ 2000 cells.

1.2 Governing Equations

The governing equations are compressible Euler with ratio of specific heats $\gamma = 1.4$. You should implement your finite volume method using the conservative state vector $\mathbf{u} = [\rho, \rho u, \rho v, \rho E]^T$. Definitions of some key quantities are as follows:

- ρ_0 : inlet stagnation density
- a_0 : inlet stagnation speed of sound
- $p_0 = \rho_0 a_0^2 / \gamma$: inlet stagnation pressure
- α : inlet angle of attack, set to 50°
- $p_{\text{out}} = 0.7p_0$: outflow static pressure

Use an inviscid wall boundary condition for the blade surface, a subsonic inflow, and a subsonic outflow. Convenient units are recommended, e.g. $\rho_0 = 1$ and $a_0 = 1$, so that state vector entries are $\mathcal{O}(1)$. All boundary conditions should be implemented according to the methods described in the notes.

1.3 Flux Functions

Implement two Riemann solvers: Roe and one of your choice (e.g. HLLE, HLLC, Osher, flux vector splitting, etc. – see *Riemann Solvers and Numerical Methods for Fluid Dynamics* by E. F. Toro). For the Roe flux, use an entropy fix with $\epsilon = 0.1c$.

1.4 Second Order and Limiters

Implement both first-order and second-order finite-volume methods. For second-order, try both no limiting and one or more of Barth-Jespersen (BJ) or the maximum-principle (MP) region limiters (e.g. LCD, Durlofsky, MLG). The choice of gradient calculation for the BJ limiter is up to you.

1.5 Steady-State Simulations

For steady-state simulations, march to steady state on a given mesh and monitor the L_1 norm of the discrete residual vector. Terminate your calculation when this norm drops five orders of magnitude (10^{-5}) below the starting residual norm. Initialize the flow to a uniform state obtained using the inlet conditions and a guessed Mach number of 0.1. For uniform-refinement studies, you can use the solution from a coarser mesh to initialize the state on a finer mesh. For the second-order solutions, you can use the first-order solutions as initial guesses.

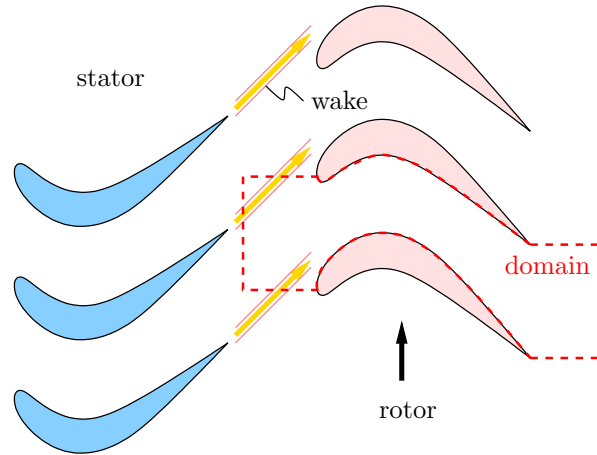
1.6 Unsteady Simulations

In addition to steady-state, of interest is also the unsteady case where the row of blades is situated behind a stator row. The motion of the blades relative to the stator creates an unsteady inflow

condition on the blades, as shown below, which you can model as a deficit in total density:

$$\begin{aligned}\rho_0(\eta(t)) &= \rho_0 [1 - f_{\text{wake}} \exp(-\eta^2/(2\delta^2))] \\ \eta(t) &= \frac{y_{\text{stator}}}{\Delta y} - \left\lfloor \frac{y_{\text{stator}}}{\Delta y} \right\rfloor - \frac{1}{2} \\ y_{\text{stator}} &= y_{\text{rot}} + V_{\text{rot}} t\end{aligned}$$

where $f_{\text{wake}} = 0.1$, $\delta = 0.1$, $\Delta y = 18\text{mm}$, y_{rot} is the y position on the rotor (i.e. the y position in your domain), V_{rot} is the rotor speed, and $\lfloor \cdot \rfloor$ is the greatest-integer (floor) function. Set the rotor speed to the stagnation speed of sound at the inflow, $V_{\text{rot}} = a_{0,\text{inflow}}$. Start the unsteady simulations with a steady-state solution and run sufficiently long until you observe periodic behavior of the outputs.



1.7 Time Stepping

Use SSP-RK2 or SSP-RK3 for time integration. In the steady-state cases, use local time stepping to accelerate convergence. For both steady and unsteady simulations, set the time step using the CFL condition.

1.8 Outputs

Your post-processing capabilities should consist of:

1. Convergence of the L_1 residual norm (log scale) versus the number of iterations.
2. Mach number and entropy field plots over the entire domain.
3. Distributions of the pressure coefficient on the blade surface (upper and lower together). The pressure coefficient c_p is given by

$$c_p = \frac{p - p_{\text{out}}}{q_{\text{out}}}, \quad \text{where } q_{\text{out}} = \frac{1}{2} \gamma p_{\text{out}} M_{\text{out}}^2, \quad M_{\text{out}}^2 = \frac{2}{\gamma - 1} \left[\left(\frac{p_0}{p_{\text{out}}} \right)^{(\gamma-1)/\gamma} - 1 \right]$$

4. A table showing convergence of the x and y force coefficients, c_x and c_y , which are the forces F_x and F_y normalized by $q_{\text{out}} c$ with $c = 18.804\text{mm}$. Calculate convergence rates relative to “truth” values of c_x and c_y from a simulation on the finest mesh you can run.

2 Tasks

Address the following tasks in your report.

1. [15%] Flux function implementation and testing. State the flux functions that you chose, describe how you verified that your flux functions are correct, and present results from one or two of your verification tests.
2. [15%] First-order solver. Describe your algorithm and include a representative c_p and Mach number plot for the steady-state case, using your choice of mesh and flux function.
3. [20%] Second-order solver and limiters. Describe your algorithm and explain your gradient calculation and limiter. Include a c_p plot for the same case as the first-order solver – you can overlay your c_p plots.
4. [20%] Convergence studies at steady-state. You do not have to run all of the possible combinations of solvers, fluxes, limiters, grids. Instead, your goal should be to run enough simulations to answer the following questions:
 - How do the flux functions differ in dissipation?
 - Is the first/second-order method actually first/second order accurate?
 - Do limiters impact the convergence rates and error magnitudes?

Use the force coefficients as the scalar outputs in your studies, and show overlaid c_p distributions to support your discussion.

5. [20%] Unsteady runs. Run simulations at both first and second order, on at least two meshes: coarse and fine, for a sufficiently long time horizon to observe several cycles of periodic output behavior. Show a few entropy field plots at various times and time histories of the force coefficients (overlay some of these to make comparisons). Discuss the effect of mesh resolution and order on the unsteady results.

Note that accompanying this assignment are several codes that you may find useful, including a Matlab Euler solver and some plotting functions.

3 Deliverables

Each group should turn in, electronically via Canvas,

1. A technical report, **12-pages maximum** for this project, as a **.pdf** file that describes your methods and results, and that answers the questions. The report should be professional, complete, and concise. **10%** of your grade will be determined by the presentation: neatness, labeling of axes, spelling, etc.
2. Documented source files of all codes you wrote for this project, as one **.zip** archive. You should provide all files necessary to run your program(s).

This is a group assignment. At the end of the report, summarize the contributions of each group member to the method development, code, results, and report.